

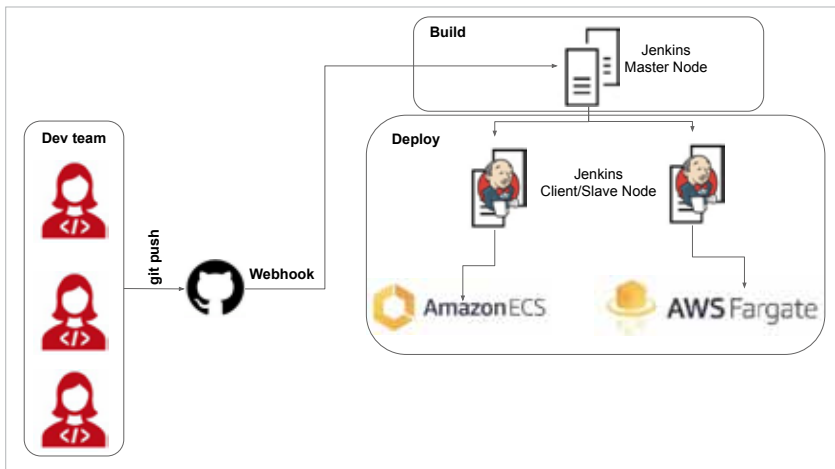


Figure 5: CloudOps using Jenkins for AWS platform

and cloud platform services are as per the geographic, industry or customer expectations.

CloudOps aims at defining a framework with the above key pillars in order to handle governance, operational activity and cloud platform management efficiently.

Automation of alerts and events in CloudOps helps to simplify activities. Azure provides Azure Monitor, AWS provides CloudWatch and GCP provides Google Cloud Monitoring to proactively monitor any service or resource for performance and health status checks. They create alerts based on a predefined rule about the threshold of the cloud service utilisation. There are other native services, too, which are very handy for the Ops team to check the health status of the cloud service and its underlying resources from a customisable dashboard facility.

Real-time monitoring and tracking of resources plays an important role in cloud architecture as it helps to understand the behaviour, health status and utilisation of cloud services easily, and helps to drive CloudOps.

## Integrating Ansible with CloudOps

Developing an integrated CloudOps pipeline activity includes continuous development, continuous build/integration and continuous deployment.

In advanced implementation, we can integrate instance provisioning, role-based permissions and various configuration modules together.

Ansible is a powerful tool for automation of infrastructure services for CloudOps integration using playbooks. Ansible playbooks with integrated CloudOps have a structure as shown in the script below:

```
Directory structure for ansible:
|--aws-ansible/azure-ansible (Playbook for AWS/Azure)
|--roles
|--instance_provisioning
|--tasks (VPC, S3, EC2 instances for AWS and VPN, VM instances for Azure)
|--main.yml
|--vars (configuration module)
|--main.yml
|--deploy.yml
```

```
|--hosts (whitelisting IP)
```

In general, for CloudOps activities, Ansible is used for orchestration of application packages to artifacts, and a build tool like Jenkins or native code pipeline can be used to build/deploy on top of cloud orchestration.

Ansible works well with AWS and Azure as an integrated orchestration and infrastructure provisioning tool, and can be integrated with native pipelines as well.

## Integrating Jenkins with CloudOps

Jenkins is a powerful CI/CD implementation tool/framework that can be used for CloudOps for build and deployment of application services. It can be integrated to AWS, Azure or GCP and be used for native integration to build and deploy Web applications, serverless applications, REST API services, and container services like Kubernetes service orchestration with Dockerized images.

During pipeline integration, Jenkins offers a facility to promote or demote an application build to various stages like development, test, UAT and production. We can use Helm for image registry, or use any container registry like ECS in AWS platform, AKS in the Azure platform, or GKE in the Google platform. We can use native monitoring tools and dashboard facilities from Jenkins to monitor and validate the builds from it. **END** 🐧

## References

- [1] [www.ansible.com](http://www.ansible.com)
- [2] [www.jenkins.io](http://www.jenkins.io)
- [3] <https://www.opcito.com/blogs/ansible-manage-aws-infrastructure>
- [4] <https://www.jenkins.io/blog/2018/08/06/serverless-cicd-jenkins/>

## By: Dr Anand Nayyar and Dr Magesh Kasthuri

Dr Anand Nayyar is a PhD in wireless sensor networks and swarms intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning and cyber security.

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.